The Brakes on the Unwinding: Federal Renewal-Compliance Enforcement and Medicaid Coverage Retention

Abstract

Background. During the 2023–2024 Medicaid continuous coverage unwinding, approximately 69 percent of coverage terminations were for procedural reasons rather than ineligibility determinations. In August 2023, the Centers for Medicare and Medicaid Services (CMS) identified a widespread compliance failure: many states were conducting automatic renewals at the household level rather than the federally required individual level, resulting in the inappropriate disenrollment of over 400,000 individuals. CMS then documented a set of state pause and delay actions while corrective work proceeded. No prior study has evaluated this enforcement episode using state-month difference-in-differences methods.

Methods. I constructed a balanced state-month panel (51 jurisdictions, 18 months, February 2023–July 2024) using publicly available CMS administrative data. I estimated the association between CMS-concurred pauses in procedural disenrollments and five renewal outcomes using two-way fixed effects (TWFE) difference-in-differences and the Gardner (2022) two-stage estimator (DID2S), exploiting staggered pause timing across 15 states. I assessed robustness through pre-trend tests, placebo simulations, Rambachan-Roth sensitivity analysis, Oster bounds, multiple-testing adjustment, approximate power calculations, early-pause exclusions, trimmed event windows, an ITT decomposition, and pause spell and intensity analyses.

Results. CMS-concurred pauses were associated with a 6.5-percentage-point increase in renewal completion rates (95% CI: 0.0 to 13.0 pp; $p = 0.05$), consistent across estimators (DID2S: +6.4 pp, $p = 0.07$). Point estimates for procedural termination rates (-5.6 pp, $p = 0.14$) and total disenrollment rates (-4.4 pp, $p = 0.11$) were directionally consistent with reduced coverage loss but did not achieve conventional significance. The ITT specification was null, renewal-completion event studies failed the baseline formal pre-trend test, the headline estimate did not survive multiple-testing adjustment, and approximate power was limited. Spell-based analyses suggested that renewal completion increased at pause onset and during active pauses without a clear post-pause catch-up spike, while longer planned pauses were associated with larger reductions in procedural terminations and total disenrollment.

Conclusions. CMS-concurred pauses were associated with higher renewal completion in pause states, but the evidence is fragile and does not support strong causal claims about federal enforcement effects. The paper is most informative as a quantitative benchmark for a rare enforcement episode and as evidence that stronger administrative data would be needed to estimate these effects credibly.

1. Introduction

The resumption of Medicaid eligibility redeterminations in 2023—following three years of federally mandated continuous coverage during the COVID-19 public health emergency—produced one of the largest coverage disruptions in the program's history. Between April 2023 and mid-2024, over 25 million individuals were disenrolled from Medicaid, with approximately 69 percent of terminations occurring for procedural reasons: beneficiaries lost coverage not because they were determined ineligible but because they failed to complete required renewal paperwork (KFF, 2025; MACPAC, 2024). This mass procedural disenrollment highlighted the central role of administrative burden in Medicaid coverage retention (Moynihan, Herd, and Harvey, 2015; Herd and Moynihan, 2018) and raised urgent questions about the capacity of federal oversight to protect beneficiaries during periods of large-scale administrative disruption.

In August 2023, CMS identified a specific compliance failure with potentially far-reaching consequences: many states were conducting automatic (ex parte) renewals at the household level rather than at the federally required individual level. Under household-level processing, if one household member was found ineligible or failed to respond to a renewal notice, other individually eligible household members could be procedurally terminated. CMS required all states to assess their compliance, and by September 21, 2023, had publicly classified 30 states as noncompliant or still assessing. Fifteen states subsequently obtained CMS concurrence to pause procedural disenrollments while implementing corrective actions, and over 400,000 erroneously terminated individuals were reinstated by January 2024 (GAO, 2024; CMS, 2024).

This federal enforcement action is distinctive in the context of Medicaid federalism. Historically, the federal-state relationship in Medicaid has relied on cooperative engagement rather than punitive enforcement (Gusmano and Thompson, 2025). The August–September 2023 compliance intervention therefore represents an unusual case of direct federal pressure that compelled immediate operational changes at the state level. At the same time, describing the episode as quasi-random would be too strong: pause states differed from non-pause states, several pauses predated the September 2023 assessment snapshot, and broader noncompliance status did not map cleanly onto later outcome changes. The episode is best viewed as a policy-relevant setting for quasi-experimental analysis rather than a clean natural experiment.

Despite a rapidly growing literature on the Medicaid unwinding, no existing study has evaluated this CMS compliance episode using state-month difference-in-differences methods. Prior causal work has focused on the unwinding itself, estimating the effect of resuming redeterminations on coverage and economic well-being (Dasgupta, 2026), or on collaborative federal interventions such as the CMS/USDS technical assistance effort to increase automated renewals in four states (Moynihan, Herd, and Ribeiro, 2025). The administrative burden literature has established that renewal processes are a critical determinant of coverage retention (Myerson, Espeseth, and Dague, 2025; Arbogast, Chorniy, and Currie, 2024), but little evidence exists on whether a federal intervention that directly altered the procedural termination pipeline changed state-level renewal outcomes. The federalism literature on Medicaid enforcement remains largely qualitative (Gusmano and Thompson, 2025).

This paper provides a structured empirical assessment of CMS individual-level renewal compliance enforcement on Medicaid coverage outcomes. Using a state-month panel constructed from publicly available CMS administrative data, I estimate difference-in-differences models that exploit the staggered timing of CMS-concurred pauses in procedural disenrollments across 15 states. My contribution is therefore narrower than a clean causal claim. I document effect sizes consistent with improved renewal processing, I show how those estimates vary when the intervention is treated as a temporary pause rather than a permanent treatment, and I clarify why the evidence remains sensitive to identification and power concerns.

I find that CMS-concurred pauses were associated with a 6.5-percentage-point increase in renewal completion rates, with directionally similar but imprecise reductions in procedural termination and total disenrollment. The broader ITT specification is null, however, and the headline association weakens materially under several diagnostics, including formal pre-trend tests, multiple-testing adjustment, and approximate power calculations. The findings are therefore best interpreted as suggestive evidence on a rare federal enforcement episode rather than definitive evidence of causal federal oversight effects.

2. Background

2.1 The Medicaid Unwinding

The Families First Coronavirus Response Act (FFCRA) of 2020 conditioned enhanced federal Medicaid matching funds on a continuous coverage provision (CCP) that prohibited states from disenrolling beneficiaries during the public health emergency. This provision increased total Medicaid enrollment from 71 million in March 2020 to over 91 million by September 2022 (Dague and Ukert, 2023). When the Consolidated Appropriations Act (CAA) of 2023 decoupled the CCP from the PHE declaration, states began resuming eligibility redeterminations starting in April 2023.

The scale of disenrollment during the unwinding was unprecedented. Over 25 million individuals were disenrolled between April 2023 and mid-2024, while approximately 56 million had coverage renewed (KFF, 2025). Approximately 69 percent of all terminations were for procedural reasons (MACPAC, 2024; GAO, 2025). Procedural termination rates varied enormously across states, from 22 percent in Maine to 93 percent in Nevada (KFF, 2025). Children were disproportionately affected: 4.16 million fewer children were enrolled by December 2023, a 10 percent national decline (Georgetown CCF, 2024).

2.2 Administrative Burden and Medicaid Renewal

The administrative burden framework conceptualizes the costs citizens face in interacting with government programs along three dimensions: learning costs, compliance costs, and psychological costs (Moynihan, Herd, and Harvey, 2015). Herd and Moynihan (2018) demonstrated that burden levels are a product of deliberate political choice rather than administrative accident. The Medicaid renewal process imposes substantial compliance costs that fall disproportionately on populations with fewer resources to manage them (Haeder et al., 2025).

Recent empirical work confirms the magnitude of these effects. Arbogast, Chorniy, and Currie (2024) found that state regulations increasing administrative burdens reduced child Medicaid enrollment by 5.9 percent within six months. Myerson, Espeseth, and Dague (2025) demonstrated through a large-scale RCT that personalized navigator assistance increased renewal by 1 percentage point overall, with substantially larger effects for tribal members (8 pp) and children (3 pp). Moynihan, Herd, and Ribeiro (2025) evaluated a federal technical assistance intervention to increase automated renewals in four states, finding that the intervention increased ex parte renewals by 21.6 percentage points and decreased procedural denials by 8.3 points.

2.3 The Individual-Level Renewal Compliance Shock

On August 30, 2023, CMS issued a State Medicaid Director Letter alerting states to a potential eligibility system issue: some states were conducting ex parte renewal processes at the household level rather than the federally required individual level (CMS, 2023). Under household-level processing, if one household member was found ineligible or failed to respond, other household members who might remain individually eligible could be procedurally terminated. By September 21, 2023, CMS published a state assessment table classifying each state's compliance status: 24 states and territories attested to conducting individual-level renewals, while 29 states (including DC) were identified as noncompliant or still assessing (CMS, 2023).

Fifteen states subsequently obtained CMS concurrence to delay procedural disenrollments, with pause start dates ranging from May 2023 (Delaware) through October 2023 (Texas). By January 2024, all identified individuals had been reinstated (CMS, 2024). The Consolidated Appropriations Act of 2023 also granted CMS new enforcement authorities, codified in an interim final rule (CMS-2023-26640), including the power to require corrective action plans, suspend procedural disenrollments, and impose civil money penalties of up to $100,000 per day.

2.4 Contribution of This Paper

This paper fills three gaps in the literature. First, it extends the federalism and enforcement literature from qualitative description (Gusmano and Thompson, 2025) to quantitative analysis of a concrete Medicaid compliance episode. Second, it advances the administrative burden literature by estimating the association between a specific procedural intervention—pausing procedural disenrollments—and renewal outcomes, complementing the existing evidence on navigator assistance (Myerson, Espeseth, and Dague, 2025) and automated renewals (Moynihan, Herd, and Ribeiro, 2025). Third, it reframes the intervention as a temporary pause with heterogeneous duration and scope, which makes it possible to study onset, active-pause, and post-pause patterns rather than relying solely on a single binary treatment indicator.

3. Data

3.1 Data Sources

I construct a balanced state-month panel of Medicaid unwinding outcomes and federal compliance enforcement status for 51 jurisdictions (50 states and the District of Columbia) spanning February 2023 through July 2024. My analysis relies on four publicly available data sources.

CMS Unwinding Renewal Metrics. My primary outcome data come from the Medicaid and CHIP Unwinding Monthly Reports, publicly available through the data.medicaid.gov portal (Dataset ID: 5abea2e0-3f8e-4b49-a50d-d63d5fd9103c). Under Section 1902(tt)(1) of the Social Security Act, added by the Consolidated Appropriations Act of 2023, states were required to submit monthly data to CMS for each month from April 2023 through June 2024 on renewals initiated and completed, disenrollment outcomes, and other operational measures. The dataset reports, for each state-month, the number of beneficiaries with a renewal due; total renewals completed (decomposed into ex parte and form-based); total disenrollments at renewal (decomposed into beneficiaries determined ineligible and those terminated for procedural reasons); and renewals still pending at month's end. I use the "Updated" data version when available, which reflects the disposition of previously pending renewals as of three months after the original reporting period, providing a more complete picture of renewal outcomes.

CMS Eligibility and Enrollment Performance Indicators. I obtain monthly state-level Medicaid and CHIP enrollment totals and operational metrics from the Performance Indicator (PI) data collection (Dataset ID: 6165f45b-ca93-5bb5-9d06-db29c692a360). This data series, submitted monthly by all states since 2013, provides point-in-time enrollment counts (total, Medicaid-only, CHIP, child, and adult), call center operations (volume, wait times, abandonment rates), and application processing metrics. I use these data both as outcome variables (enrollment levels) and to compute per-capita-normalized outcome measures.

CMS State Compliance Assessment. My primary treatment variable derives from the "Preliminary Overview of State Assessments Regarding Compliance with Medicaid and CHIP Automatic Renewal Requirements at the Individual Level," published by CMS on September 21, 2023 (updated September 28, 2023). Following an August 30, 2023 State Medicaid Director Letter alerting states to a potential eligibility system issue—specifically, that some states were conducting ex parte renewals at the household level rather than the federally required individual level—CMS required all states to assess and report their compliance status. The resulting table classified each state as either (a) attesting to correctly conducting auto-renewals at the individual level (24 states and territories), or (b) not conducting individual-level auto-renewals or still assessing (29 states including DC). For states in the latter group, the table further documented affected populations (children, household members with different eligibility statuses) and estimated numbers of affected individuals.

CMS State Option to Delay Procedural Disenrollments. My secondary (as-treated) treatment variable comes from CMS's publicly available chart documenting 15 states that obtained CMS concurrence to delay procedural disenrollments for one or more months (chart as of May 22, 2024). This chart records, for each state, the duration of the delay, the months in which the delay was active, the scope (all beneficiaries vs. specific populations), and the affected populations. These delays are the main documented state-level operational response to the compliance episode and vary meaningfully in duration, scope, and population coverage.

American Community Survey. I supplement the CMS administrative data with state-level demographic controls from the 2022 American Community Survey 1-Year Estimates, accessed via the Census Bureau API. I include total population, percent Black, percent Hispanic, percent below the federal poverty line, and median household income as time-invariant state-level covariates.

3.2 Treatment Definition

I define treatment using two complementary approaches. My intent-to-treat (ITT) specification codes a binary indicator equal to one for the 30 states classified as noncompliant or still assessing in the September 21, 2023 CMS assessment, interacted with a post-assessment indicator (months from September 2023 onward). The 21 states attesting to compliance serve as the comparison group. This ITT approach captures the effect of being flagged by CMS for noncompliance, regardless of whether the state subsequently paused disenrollments.

My as-treated specification exploits the staggered timing of pause and delay implementation. Fifteen states obtained CMS concurrence to delay procedural disenrollments, with start dates ranging from May 2023 (Delaware) through October 2023 (Texas). A binary pause indicator turns on in the first month of the CMS-concurred delay and remains on through the end of the delay period. Because several pauses began before the September 2023 assessment snapshot and because pause duration and scope varied across states, I also report sensitivities that exclude early-pause states and specifications that decompose pause spells and intensity.

3.3 Sample Construction

My analysis panel consists of 918 state-month observations (51 jurisdictions x 18 months). The panel is balanced in terms of the state-month skeleton, though outcome variables are missing for approximately 11 percent of observations, concentrated in the earliest months (February–April 2023) when many states had not yet initiated unwinding-related renewals. This missingness is structural:

states that initiated unwinding later in the spring report zero renewals due in earlier months. I treat these as structurally missing rather than imputing values. For the primary analysis, I restrict the sample to months in which a state has non-missing renewal data, yielding 764 state-month observations with complete outcome data.

3.4 Key Outcome Variables

My primary outcomes are:

1. **Procedural termination rate**: the share of total disenrollments at renewal due to procedural (rather than eligibility) reasons, measuring the intensity of administrative burden in the renewal process.
2. **Total disenrollment rate**: total disenrollments divided by renewals due, capturing the overall rate of coverage loss at renewal.
3. **Renewal completion rate**: renewals completed divided by renewals due, measuring the share of beneficiaries who successfully retained coverage through the renewal process.
4. **Ex parte renewal rate**: the share of completed renewals conducted on an ex parte (automated) basis, reflecting the degree to which states successfully automate the renewal process.
5. **Pending renewal rate**: renewals still pending at month's end as a share of renewals due, capturing administrative backlogs.

3.5 Descriptive Statistics

Table 1 presents pre-assessment summary statistics for key variables by treatment group. Prior to the September 2023 compliance assessment, noncompliant states had modestly lower procedural termination rates (65.5% vs. 74.2%, normalized difference = -0.46) and lower disenrollment rates (35.1% vs. 39.9%) than compliant states. Noncompliant states also tended to be smaller (mean enrollment 1.42 million vs. 2.41 million) and had higher median household incomes ($78,388 vs. $68,813, normalized difference = 0.85). These level differences motivate the inclusion of state fixed effects, which absorb all time-invariant state characteristics. The relevant identification assumption is that outcome *trends* would have been parallel absent treatment, which I evaluate through pre-trend tests.

4. Methods

4.1 Study Design

I use a difference-in-differences (DiD) design to estimate the association between CMS-required renewal compliance actions and Medicaid procedural termination rates and related outcomes during the continuous coverage unwinding period (April 2023–July 2024). The variation of interest comes from CMS's September 21, 2023 compliance assessment, which identified 30 states as noncompliant or still assessing their individual-level automatic renewal practices, and from the staggered timing of CMS-concurred pauses and delays in procedural disenrollments across 15 states. I treat this as a policy-relevant quasi-experimental setting rather than a clean natural experiment.

4.2 Estimating Equations

Two-way fixed effects (TWFE). My baseline specification is:

$$Y_{st} = \alpha_s + \delta_t + \beta \cdot \text{Treatment}_{st} + \varepsilon_{st}$$

where Y_{st} is the outcome for state s in month t, α_s are state fixed effects absorbing time-invariant state characteristics, δ_t are month fixed effects absorbing common temporal shocks, and Treatment_{st} is either the ITT indicator ($\text{Noncompliant}_s \times \text{Post}_t$) or the as-treated indicator (PauseActive_{st}). The coefficient β estimates the average treatment effect on the treated (ATT). Standard errors are clustered at the state level to account for serial correlation within states.

DID2S (Gardner 2022). To address potential bias from treatment effect heterogeneity in staggered DiD settings, I implement the two-stage estimator of Gardner (2022). In the first stage, I estimate the state and month fixed effects using only untreated observations (state-months where no pause/delay is active). In the second stage, I regress the residualized outcome on the treatment indicator using all observations with state-level clustered standard errors.

Event study. To assess the parallel trends assumption and characterize dynamic treatment effects, I estimate:

$$Y_{st} = \alpha_s + \delta_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - T_s^* = k] + \varepsilon_{st}$$

where T_s^* is the month state s's pause began and k indexes event time in months relative to pause onset, normalized to $k = -1$. I restrict the event window to $k \in [-6, +12]$ and estimate this specification on the 15 pause/delay states only. Pre-treatment coefficients $\{\beta_k : k < -1\}$ provide a visual and statistical test of parallel trends.

4.3 Robustness and Sensitivity Analyses

I conduct a broad set of robustness and sensitivity checks. These include: (1) formal pre-trend tests using joint Wald statistics on pre-treatment event study coefficients; (2) placebo simulations with 500 iterations randomly assigning pseudo-pause dates to control states; (3) Rambachan and Roth (2023) sensitivity analysis computing confidence intervals under varying degrees of parallel-trends violations; (4) a balanced-panel restriction (months with at least 90% state reporting coverage); (5) alternative treatment definitions, including an absorbing-treatment sensitivity and the ITT specification with demographic controls; (6) heterogeneity analysis exploiting whether children were flagged as affected; (7) a mechanisms test examining whether procedural terminations and pending renewals move in predicted directions; (8) Oster (2019) bounds for sensitivity to selection on unobservables; (9) comparison between the full and restricted panels to assess sensitivity to structural missingness; (10) a sensitivity excluding the 10 pause states whose pauses began before September 2023; (11) multiple-testing adjustments across the 15 main treatment-effect estimates; (12) an approximate minimum detectable effect calculation for the as-treated design; (13) trimmed event-window event studies using k in [-4, +12]; (14) a decomposition of the null ITT result based on overlap between ITT-treated and pause states; and (15) pause spell and intensity specifications that separate onset, during-pause, and post-pause periods and allow treatment effects to vary with duration and scope. Because the intervention is temporary rather than absorbing, estimators designed for permanent treatment adoption, such as Sun-Abraham or Callaway-Sant'Anna, are not a clean substitute for the main specification without stronger assumptions about post-pause exposure.

4.4 Software

All analyses are implemented in Python using pyfixest for fixed-effects estimation, statsmodels for the DID2S two-stage estimator with cluster-robust variance estimation, and matplotlib/seaborn for visualization.

5. Results

5.1 Main Results

Table 2 presents the main difference-in-differences results across all five outcomes, comparing intent-to-treat (ITT), as-treated TWFE, and DID2S estimators.

Renewal completion rate. The as-treated TWFE specification yields a coefficient of +6.5 percentage points (SE = 3.2 pp, p = 0.05), indicating that states with active CMS-concurred pauses experienced higher renewal completion rates relative to states without pauses (Table 2, column 2). The DID2S estimator produces a nearly identical point estimate of +6.4 percentage points (SE = 3.6 pp, p = 0.07). By contrast, the ITT specification, which compares all 30 noncompliant states to the 21 compliant states regardless of whether they actually paused disenrollments, yields a null estimate (-0.04 pp, p = 0.99). A simple decomposition helps explain this disconnect: only 10 of the 15 pause states were in the ITT-treated group, while 5 pause states were classified as compliant, and the 20 ITT-treated states without pauses showed virtually no post-assessment change in renewal completion.

Procedural termination rate. CMS-concurred pauses were associated with a 5.6-percentage-point reduction in the procedural termination rate (SE = 3.8 pp, p = 0.14). The direction and magnitude are consistent with the hypothesized mechanism, but the estimate is imprecise. The DID2S estimate is attenuated (-3.1 pp, p = 0.46), and the ITT estimate is positive (+3.7 pp, p = 0.37), again suggesting that noncompliance designation alone is not a good proxy for the operational treatment.

Total disenrollment rate. The as-treated estimate of -4.4 percentage points (SE = 2.7 pp, p = 0.11) is directionally consistent with reduced coverage loss at renewal, though the estimate remains below conventional significance thresholds. The DID2S estimate is -2.7 percentage points (p = 0.31).

Ex parte renewal rate. I find no detectable effect on ex parte renewal rates (+0.5 pp, SE = 2.0 pp, p = 0.81). This is consistent with the nature of the intervention: the compliance action targeted the procedural termination pipeline, not the ex parte renewal process itself.

Pending renewal rate. The point estimate of -2.1 percentage points (SE = 2.6 pp, p = 0.42) is imprecise and does not support strong conclusions about administrative backlogs. The DID2S estimate is somewhat larger (-3.8 pp, p = 0.19), but the confidence interval remains wide.

5.2 Event Study Results

Figure 2 presents event study plots for each outcome, with dynamic treatment effects plotted relative to the month before pause onset (k = -1). For the renewal completion rate, pre-treatment coefficients are generally close to zero for event times -4 through -1, though the k = -5 coefficient is elevated, contributing to a formal pre-trend rejection (Wald chi-squared(4) = 12.89, p = 0.01). The procedural termination rate shows a similar but weaker pattern (Wald test: p = 0.06), while the ex parte rate shows neither meaningful pre-trend concerns nor a post-treatment response. Figure 3 presents a multi-panel summary of all five event studies.

Trimming the event window from k in [-6, +12] to k in [-4, +12] removes the k = -5 lead and eliminates formal pre-trend rejection for renewal completion (p = 0.25 rather than 0.01) and total disenrollment (p = 0.50 rather than 0.03). Even under this narrower window, however, the period-0 coefficients remain small and imprecise. The event-study evidence therefore reduces concern about one influential early lead but does not resolve the broader identification issue.

5.3 Robustness Checks

Pre-trend, placebo, and sensitivity diagnostics. Formal Wald tests fail to reject parallel pre-trends for three of five outcomes: procedural termination rate (p = 0.06), ex parte renewal rate (p = 0.98), and pending renewal rate (p = 0.24). Pre-trend violations are detected for the total disenrollment rate (p = 0.03) and renewal completion rate (p = 0.01), driven primarily by the k = -5 event-time coefficient (Table 3). Placebo tests are supportive but not decisive: the renewal completion rate has an empirical p-value of 0.04, while the total disenrollment and procedural termination estimates fall in the tails of their placebo distributions but remain indistinguishable from noise at conventional levels.

Rambachan-Roth and Oster diagnostics. Confidence intervals widen substantially under even modest violations of parallel trends. At M = 0 (exact parallel trends imposed), the renewal completion rate confidence interval is (-0.04, 0.15). At M = 1 times the calibrated bound, no outcome remains excludable from zero (Table A2). Oster (2019) delta-star values are also small for all outcomes, including 0.033 for the renewal completion rate, indicating that relatively modest unobserved confounding could explain the estimate (Table A3).

Additional design checks. Restricting to a balanced panel with at least 90 percent state reporting coverage (N = 705) yields attenuated but directionally consistent estimates: renewal completion rate +4.9 pp (p = 0.06) and procedural termination rate -3.3 pp (p = 0.50). An absorbing-treatment sensitivity produces slightly larger coefficients, but this specification is substantively awkward because the actual intervention turns off in some states. Excluding the 10 states whose pauses began before September 2023 leaves only 5 treated states; the renewal-completion estimate remains positive (+7.4 pp in TWFE, p = 0.11; +7.3 pp in DID2S, p = 0.09) but becomes less precise.

Multiple testing, power, and mechanisms. The headline renewal-completion estimate does not survive multiple-testing adjustment across the 15 main treatment-effect tests (Benjamini-Hochberg adjusted p = 0.52; Bonferroni adjusted p = 0.75). My approximate minimum detectable effect for renewal completion is 8.05 percentage points, above the observed 6.45 percentage-point estimate, indicating limited power for effects of the size I observe. The mechanisms analysis remains directionally consistent: the procedural disenrollment-to-renewals-due rate declines by 4.6 percentage points (p = 0.04), which is the most direct measure of the intervention's intended effect (Table A4).

Other heterogeneity and sample checks. Among noncompliant states, those where children were specifically flagged as affected did not show differential effects on most outcomes, though the interaction term for the renewal completion rate is negative and marginally significant (-8.7 pp, p = 0.08), suggesting possible heterogeneity that warrants more granular data (Table A5). Results are identical between the full panel (N = 918) and the restricted panel (N = 764), as the structural missingness occurs exclusively in state-months that have no outcome data and are automatically dropped from regressions (Table A6).

5.4 Pause Spell And Intensity Analyses

The temporary nature of the intervention motivates a pause-spell decomposition. When I separate the first active month of a pause, active pause months after onset, and months after the pause ends, the renewal-completion association appears immediately at onset (+10.7 pp, $p = 0.002$) and remains positive during active pause months (+9.0 pp, $p = 0.08$; Table A17). For procedural termination and total disenrollment, onset and during-pause coefficients are negative, but most estimates are imprecise. Importantly, I do not observe a clear post-pause catch-up spike: post-pause procedural terminations remain negative (-4.2 pp, $p = 0.50$) and post-pause total disenrollment is also negative (-2.2 pp, $p = 0.67$). This pattern is more consistent with at least some prevented coverage loss than with a simple mechanical delay, although post-pause leverage is limited because many states remained paused through June 2024.

I also estimate exploratory intensity specifications that allow the active-pause effect to vary by planned delay duration and by whether the pause was narrow in scope or targeted to a subset of beneficiaries. Longer planned pauses appear materially stronger on loss outcomes: each extra planned month beyond one is associated with an 11.7-percentage-point reduction in procedural terminations ($p = 0.001$) and a 6.3-percentage-point reduction in total disenrollment ($p = 0.01$; Table A18). Narrow or targeted pauses look worse on procedural terminations (+10.4 pp relative to broader pauses, $p = 0.02$). Renewal-completion heterogeneity is near zero. Because all targeted-population pauses in this panel are also partial-scope pauses, the scope and population margins are not separately identified, so these intensity results should be treated as exploratory rather than definitive.

5.5 Log-Level Outcomes

As a semi-elasticity specification, I estimate the effect of pauses on log-transformed outcome levels. Log procedural disenrollments decline by 50.7 percent ($p = 0.19$) and log total disenrollments by 24.3 percent ($p = 0.24$), though neither is statistically distinguishable from zero. Log renewals completed shows no effect (+1.7%, $p = 0.88$). These log-level results are directionally consistent with the rate-based findings but remain imprecise due to the additional variance introduced by level differences across states (Table A7).

6. Discussion

6.1 Summary of Findings

This study provides a quantitative evaluation of CMS individual-level renewal compliance enforcement on Medicaid coverage outcomes during the 2023–2024 continuous coverage unwinding. My primary finding is that CMS-concurred pauses in procedural disenrollments were associated with a 6.5-percentage-point increase in renewal completion rates, with directionally similar but imprecise reductions in procedural termination and total disenrollment. The association is consistent across TWFE and DID2S estimators and supported by placebo tests, but it is materially weakened by the null ITT result, failed baseline pre-trend tests for the headline outcome, multiple-testing adjustment, and limited power.

6.2 Interpretation in Context

The magnitude of my renewal completion estimate (+6.5 pp) is comparable to effects found in related studies. Moynihan, Herd, and Ribeiro (2025) estimated that the CMS/USDS ex parte automation intervention increased overall renewals by 7.7 percentage points, while Myerson, Espeseth, and Dague (2025) found that personalized navigator assistance increased renewal by 1 percentage point overall, with larger effects for some subpopulations. The similarity in magnitude is notable, but unlike those settings, my evidence is materially more fragile. The null ITT estimate and the sensitivity diagnostics imply that the pause association should be read as suggestive rather than definitive.

My findings are also broadly consistent with Dasgupta's (2026) estimates of the unwinding's effect on coverage. Dasgupta found that the unwinding reduced state-level Medicaid enrollment by approximately 4 percent but did not significantly affect overall uninsurance, suggesting considerable transitions to alternative coverage sources. My positive renewal-completion estimates are consistent with a world in which the renewal process itself—not underlying eligibility—is a primary driver of coverage loss. The spell decomposition sharpens that interpretation somewhat by showing an immediate positive association at pause onset and no clear post-pause catch-up spike, though the post-pause results remain underpowered because many states stayed paused until the end of the observed period.

The null result on ex parte renewal rates is also informative. The enforcement action targeted inappropriate procedural terminations stemming from household-level processing, not the automation of the renewal process itself. The finding that procedural termination rates decline while ex parte rates remain unchanged suggests that the pause mechanism operates through added processing time and reduced inappropriate terminations rather than through more automated renewals.

6.3 Relation to Administrative Burden Literature

My results contribute to the growing empirical literature on administrative burden in health insurance programs. The administrative burden framework (Moynihan, Herd, and Harvey, 2015; Herd and Moynihan, 2018) predicts that interventions reducing compliance costs should improve program retention. The CMS compliance action reduced burden in a specific way: by halting the procedural termination of individuals who may have been inappropriately processed at the household level, it gave beneficiaries additional time and opportunity to complete the renewal process. My positive renewal-completion estimate is consistent with this mechanism, even if the evidence falls short of a definitive causal demonstration.

This finding complements the experimental evidence of Myerson, Espeseth, and Dague (2025), who demonstrated that individual-level assistance can overcome renewal barriers, and the quasi-experimental evidence of Arbogast, Chorniy, and Currie (2024), who documented the aggregate effects of administrative burden on child enrollment. My contribution is to show that federal enforcement may operate as a mechanism for burden reduction at scale—not through individual assistance or system automation, but through regulatory action that changes state administrative behavior.

6.4 Policy Implications

Three policy implications emerge, each tentative.

First, federal compliance enforcement may matter when it is translated into a concrete operational change rather than a broad compliance designation. The null ITT estimate implies that being flagged by CMS for noncompliance did not, by itself, produce detectable average changes in renewal outcomes. The action that appears most relevant is the actual pause or delay of procedural disenrollments.

Second, pause duration and scope may matter for effectiveness. The exploratory intensity results suggest that longer planned pauses were associated with larger reductions in procedural terminations and total disenrollment, while narrower or targeted pauses performed worse on procedural terminations. If those patterns generalize, policymakers should think about enforcement not only in binary terms but also in terms of how much operational relief is required to repair renewal problems.

Third, the compliance episode underscores the importance of better federal monitoring data. Without individual-level information on who was affected, paused, reinstated, and later disenrolled, even a salient enforcement action is difficult to evaluate credibly. Future federal oversight should incorporate stronger operational data collection rather than relying solely on post hoc public summaries.

6.5 Limitations

Several limitations should be considered when interpreting my findings.

Statistical power. With 51 states, 15 pause states, and 18 months, my study has limited power to detect modest treatment effects. The observed renewal-completion effect size falls below my approximate 80 percent-power minimum detectable effect, and the state-level clustering with only 51 clusters further limits precision.

Identification concerns. Formal pre-trend tests detect violations for the total disenrollment rate and the renewal completion rate under the baseline event window. Trimming the event window softens but does not eliminate these concerns. The null ITT result, small Oster bounds, and Rambachan-Roth sensitivity all point in the same direction: the headline estimate is fragile to departures from the identifying assumptions.

Multiple testing and model dependence. No main estimate survives Benjamini-Hochberg or Bonferroni adjustment. Several supportive results depend on modeling choices such as the event-study window or whether early-pause states are retained. These features do not invalidate the findings, but they limit how strongly they can be interpreted.

Treatment heterogeneity and temporary exposure. The as-treated variable collapses substantial heterogeneity in pause duration, scope, and target population into one indicator. My exploratory intensity analysis partly addresses this, but scope and population are not separately identified in the available panel. More broadly, the intervention is temporary rather than absorbing, which makes standard permanent-treatment staggered-DiD estimators an imperfect fit.

Ecological inference and limited post-pause leverage. Outcomes are measured at the state-month level rather than the individual level, so I cannot directly identify who was protected, reinstated, or later disenrolled. The spell decomposition suggests no clear post-pause catch-up spike, but post-pause leverage is weak because many states remained paused through June 2024 and contribute very few post-pause observations.

Concurrent policies. States implemented numerous mitigation strategies during the unwinding—including call center expansions, additional outreach campaigns, and extended processing timelines—that are not individually controlled for in my models. While state fixed effects absorb time-invariant differences and month fixed effects absorb common temporal shocks, state-specific time-varying confounders remain a concern.

6.6 Future Research

Several avenues for future research would strengthen the evidence base. First, individual-level analyses using T-MSIS claims or eligibility data could estimate the effect of pause policies on coverage continuity, reinstatement, health care utilization, and health outcomes for directly affected beneficiaries. Second, more granular treatment coding—incorporating the realized scope, duration, and operational intensity of state-level compliance responses—could improve estimation precision and reveal important heterogeneity. Third, longer follow-up periods would allow researchers to assess whether pause-related coverage retention persisted beyond the unwinding period or represented temporary delays in disenrollment. Fourth, the enforcement mechanisms created by the CAA of 2023 provide a policy laboratory for studying the broader question of how federal enforcement tools shape state administrative behavior in jointly administered programs.

7. Conclusion

This study provides a quantitative benchmark for a rare federal compliance-enforcement episode during the Medicaid unwinding. I find that CMS-concurred pauses in procedural disenrollments were associated with higher renewal completion in pause states, with directionally similar but imprecise reductions in procedural termination and total disenrollment. The null ITT result, failed baseline pre-trend tests for the headline outcome, unfavorable sensitivity analyses, and limited power mean these findings should be interpreted as suggestive rather than causal.

The Medicaid unwinding demonstrated that administrative processes—not only eligibility status—are a major driver of coverage loss. This episode suggests that federal enforcement may matter when it forces concrete operational changes, but it also shows the limits of what can be learned from publicly available state-month data. Stronger individual-level administrative data would be needed to determine how much federal oversight actually prevented inappropriate coverage loss.

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Tables and Figures

Table 1. Pre-Assessment Descriptive Statistics by Treatment Group

Variable	Treated Mean	Treated SD	Control Mean	Control SD	Norm. Diff.
Panel A: Renewal Outcomes					
Procedural termination rate	0.655	0.228	0.742	0.141	-0.46
Disenrollment rate	0.351	0.160	0.399	0.133	-0.33
Renewal completion rate	0.593	0.158	0.555	0.134	0.25
Ex parte renewal rate	0.476	0.225	0.549	0.258	-0.30
Pending renewal rate	0.056	0.098	0.046	0.057	0.13

Variable	Treated Mean	Treated SD	Control Mean	Control SD	Norm. Diff.
Panel B: State Characteristics					
Total enrollment (millions)	1.42	1.56	2.41	3.02	-0.41
Baseline enrollment (millions)	1.43	1.54	2.43	2.99	-0.42
Poverty rate (%)	11.2	2.1	13.3	2.5	-0.91
Median household income ($)	78,388	11,358	68,813	11,142	0.85
Black population (%)	9.5	10.0	12.7	9.9	-0.32
Hispanic population (%)	12.7	9.5	13.1	11.6	-0.05

Notes: Pre-assessment period defined as months before September 2023. Treated group: 30 states classified as noncompliant or still assessing in the CMS September 21, 2023 assessment. Control group: 21 states attesting to compliance. Normalized differences calculated as the difference in means divided by the square root of the average of the group variances. N(treated) = 126–210 state-months; N(control) = 83–147 state-months depending on variable availability.

Table 2. Main Difference-in-Differences Results

Outcome	ITT TWFE		As-Treated TWFE		DID2S	
	Coef.	p-value	Coef.	p-value	Coef.	p-value
Procedural termination rate	0.037	0.37	-0.056	0.14	-0.031	0.46
	(0.041)		(0.038)		(0.042)	
Total disenrollment rate	0.012	0.70	-0.044	0.11	-0.027	0.31
	(0.031)		(0.027)		(0.026)	
Renewal completion rate	-0.000	0.99	0.065	0.05	0.064	0.07
	(0.033)		(0.032)		(0.036)	

Outcome	ITT TWFE		As-Treated TWFE		DID2S	
Ex parte renewal rate	0.028	0.37	0.005	0.81	0.008	0.77
	(0.031)		(0.020)		(0.027)	
Pending renewal rate	-0.011	0.50	-0.021	0.42	-0.038	0.19
	(0.017)		(0.026)		(0.029)	
N	764		764		764	
State FE	Yes		Yes		Yes	
Month FE	Yes		Yes		Yes	

Notes: Standard errors clustered at the state level in parentheses. ITT treatment: noncompliant status (30 states) x post-September 2023. As-treated: `pause_active` indicator for 15 states with CMS-concurred pauses. DID2S: Gardner (2022) two-stage estimator using untreated observations for first-stage fixed effect estimation. Coefficients are reported in proportions, with p-values shown in adjacent columns.

Table 3. Summary of Robustness Checks

Test	Procedural Rate	Disenrollment Rate	Renewal Completion Rate	Ex Parte Rate	Pending Rate
Pre-trend p-value, [-6,+12]	0.06	0.03	0.01	0.98	0.24
Pre-trend p-value, [-4,+12]	0.47	0.50	0.25	0.94	0.29
Placebo empirical p-value	0.25	0.15	0.04	0.88	0.21
Balanced panel coef. (p)	-0.033 (0.50)	-0.024 (0.41)	0.049 (0.06)	-0.015 (0.44)	-0.025 (0.42)
Early-pause exclusion coef. (p)	-0.076 (0.32)	-0.018 (0.64)	0.074 (0.11)	0.018 (0.52)	-0.056 (0.22)
Absorbing treatment coef. (p)	-0.068 (0.25)	-0.047 (0.29)	0.087 (0.07)	0.015 (0.67)	-0.040 (0.27)

Test	Procedural Rate	Disenrollment Rate	Renewal Completion Rate	Ex Parte Rate	Pending Rate
Rambachan-Roth CI, M = 1	[-0.52, 0.32]	[-0.35, 0.23]	[-0.24, 0.34]	[-0.24, 0.14]	[-0.07, 0.08]
Benjamini-Hochberg p-value	0.53	0.53	0.52	0.87	0.68
Approximate MDE (pp)	12.89	8.18	8.05	17.00	5.33
Oster delta-star	-0.024	-0.040	0.033	0.002	-0.087

Notes: All estimates use the as-treated (`pause_active`) specification unless otherwise noted. Coefficients are in percentage points when applicable. Early-pause exclusion drops DC, DE, IL, ME, MI, MN, NH, NJ, OK, and SC, leaving 5 pause states. Benjamini-Hochberg p-values are applied across the 15 main treatment-effect estimates in Table 2. Rambachan-Roth confidence intervals use M = 1 x the maximum pre-period slope change. Oster delta-star represents the degree of proportional selection on unobservables, relative to observables, needed to explain away the estimated effect.

Figure 1. Timeline of CMS Compliance Enforcement and State Pause Actions

[See analysis/figures/ for timeline visualization]

Figure 2. Event Study: Dynamic Treatment Effects by Outcome

[See the replication workflow]

Panel (a): Procedural Termination Rate. Panel (b): Total Disenrollment Rate. Panel (c): Renewal Completion Rate. Panel (d): Ex Parte Renewal Rate. Panel (e): Pending Renewal Rate. Each plot shows event-time coefficients (months relative to pause onset, normalized to k = -1) with 95% confidence intervals. Dashed vertical line indicates pause onset.

Figure 3. Multi-Panel Event Study Summary

Figure 4. Placebo Test Distributions

Distribution of 500 placebo treatment effects from random assignment of pseudo-pause dates to control states. Vertical dashed line indicates the actual estimated treatment effect from the as-treated specification.

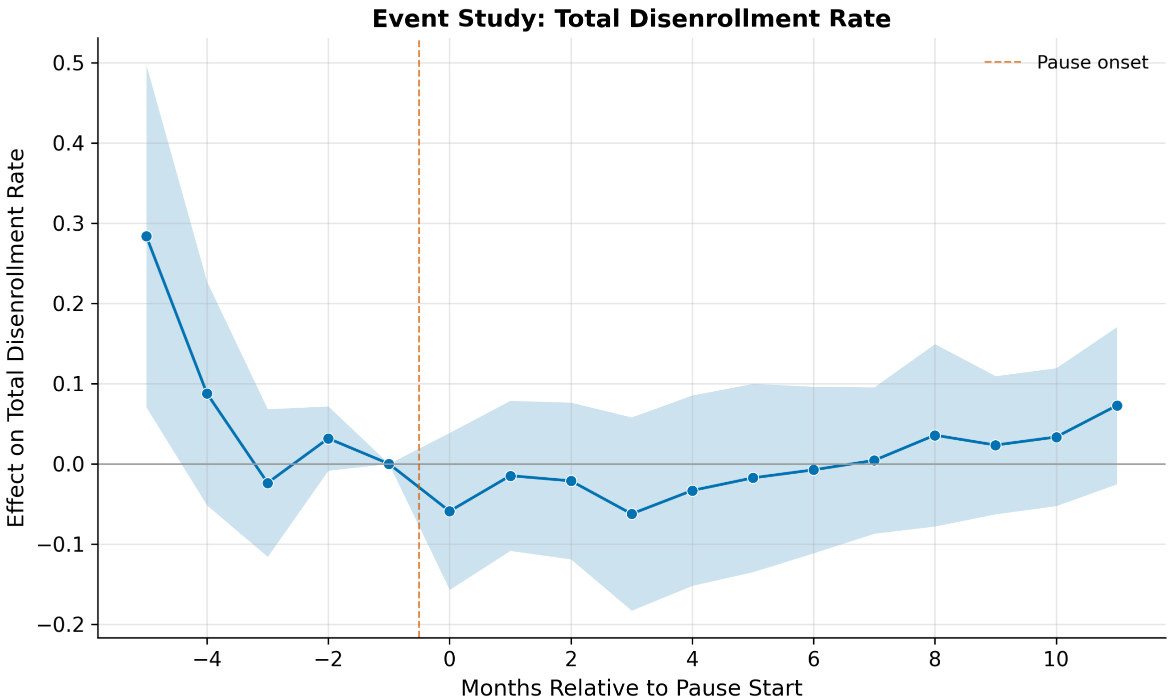

Figure 1: Fig2 Event Study Disenrollment Rate

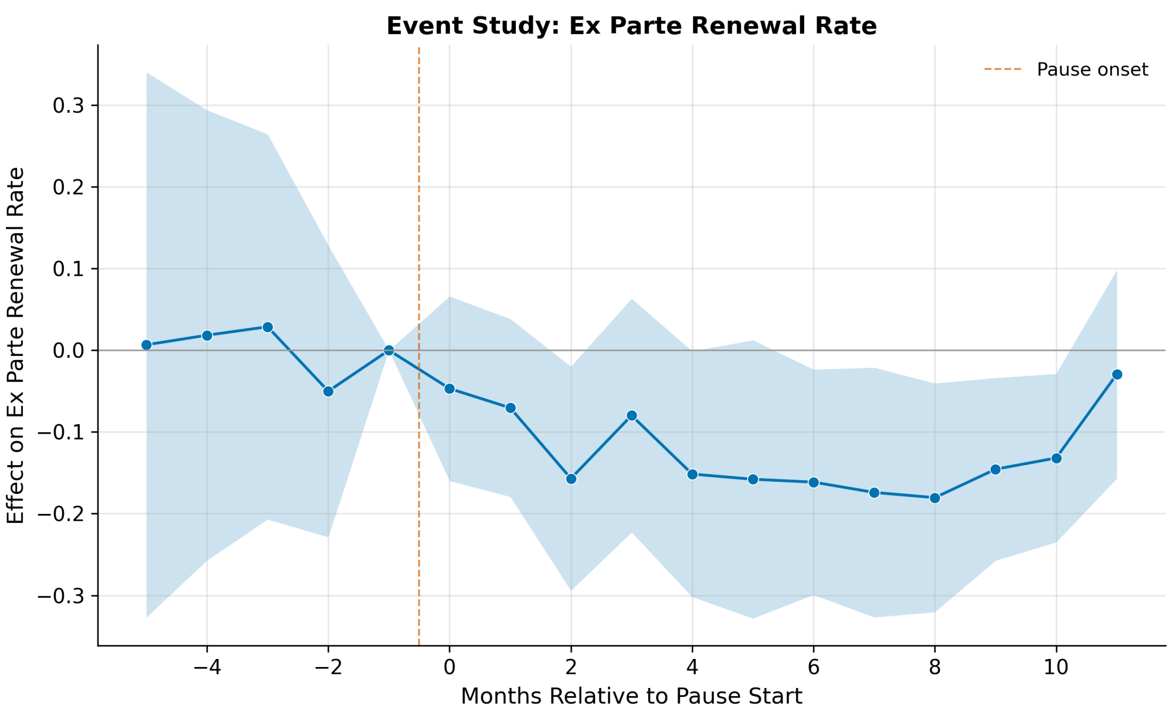

Figure 2: Fig2 Event Study Ex Parte Rate

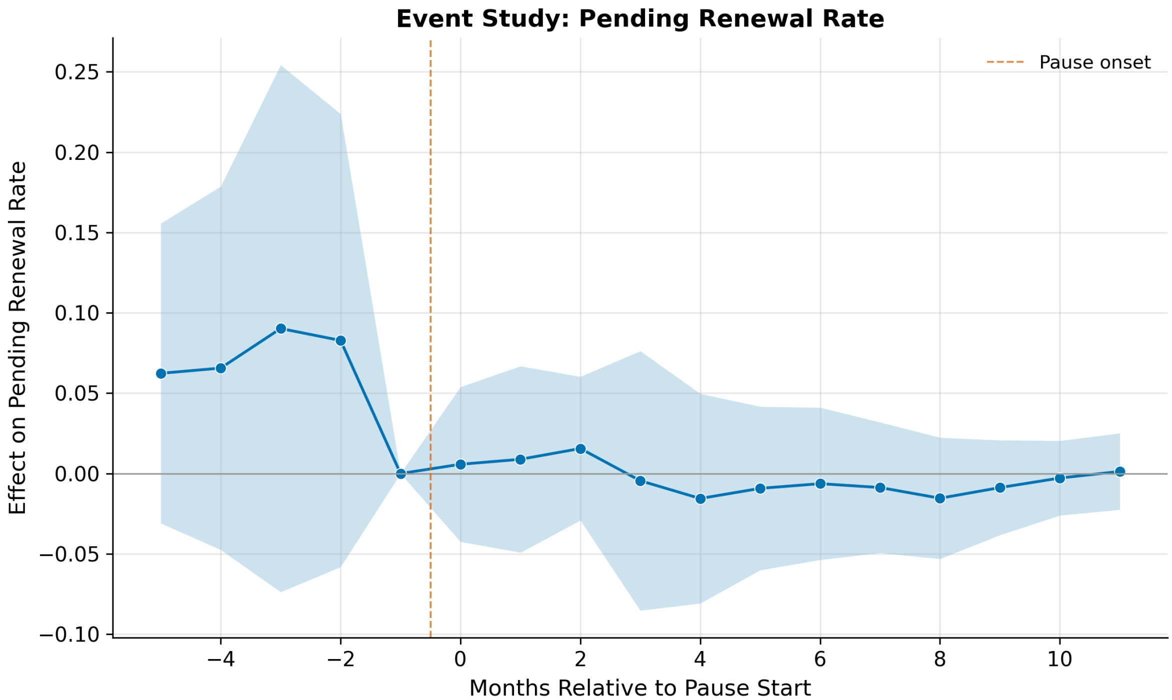

Figure 3: Fig2 Event Study Pending Rate

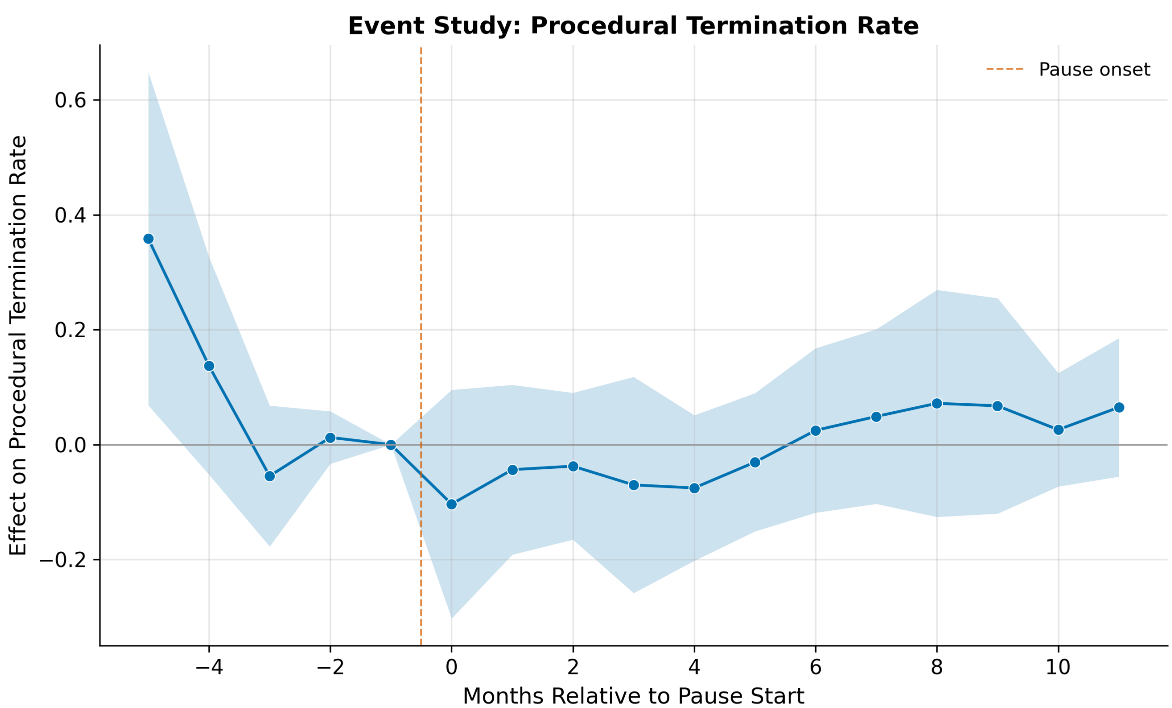

Figure 4: Fig2 Event Study Procedural Rate

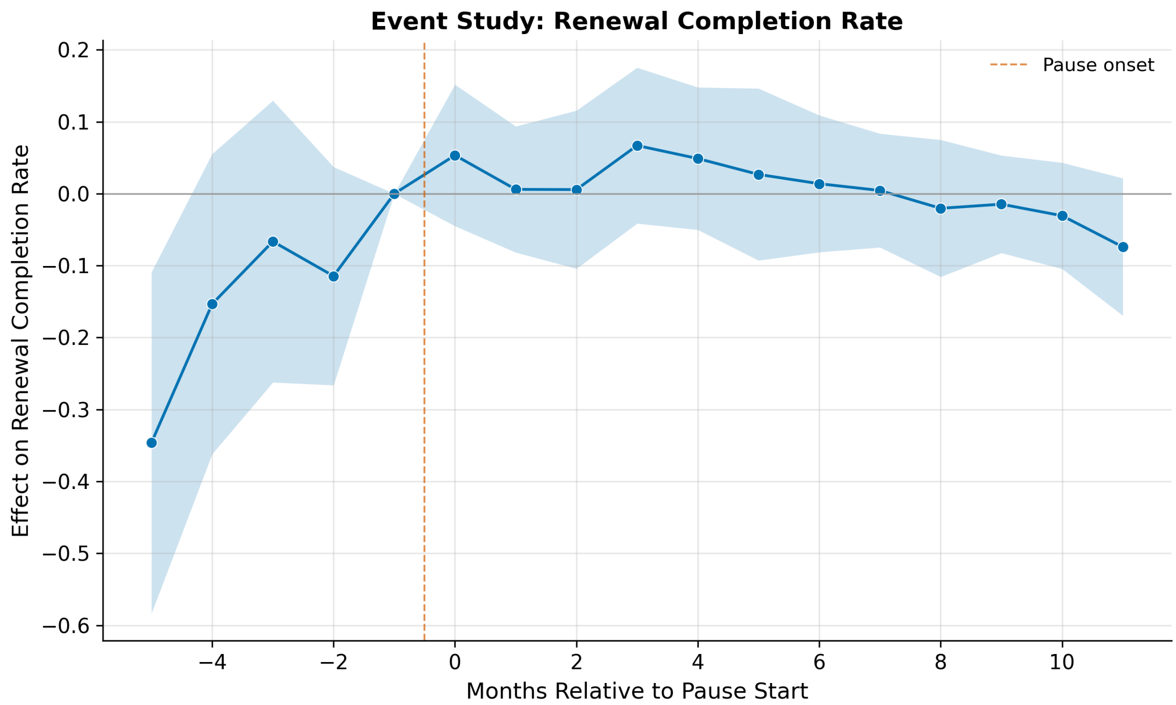

Figure 5: Fig2 Event Study Renewal Completion Rate

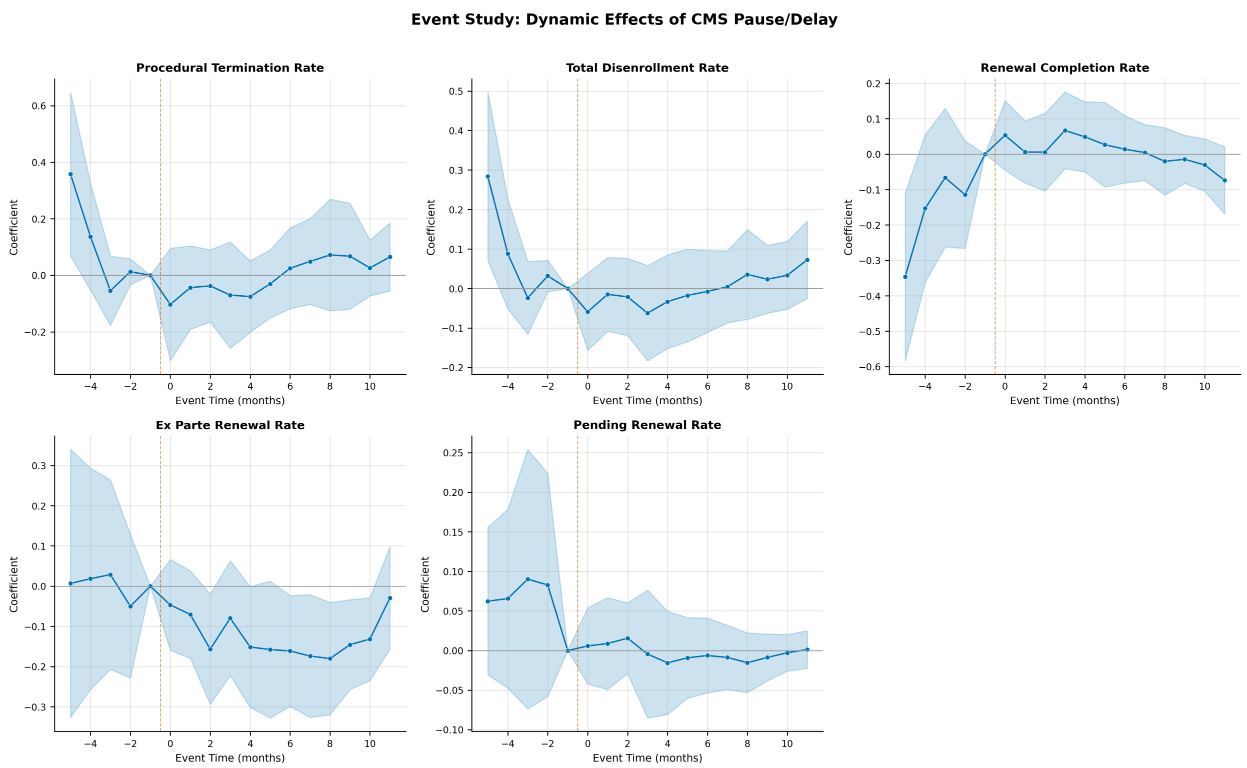

Figure 6: Fig3 Event Study Panel

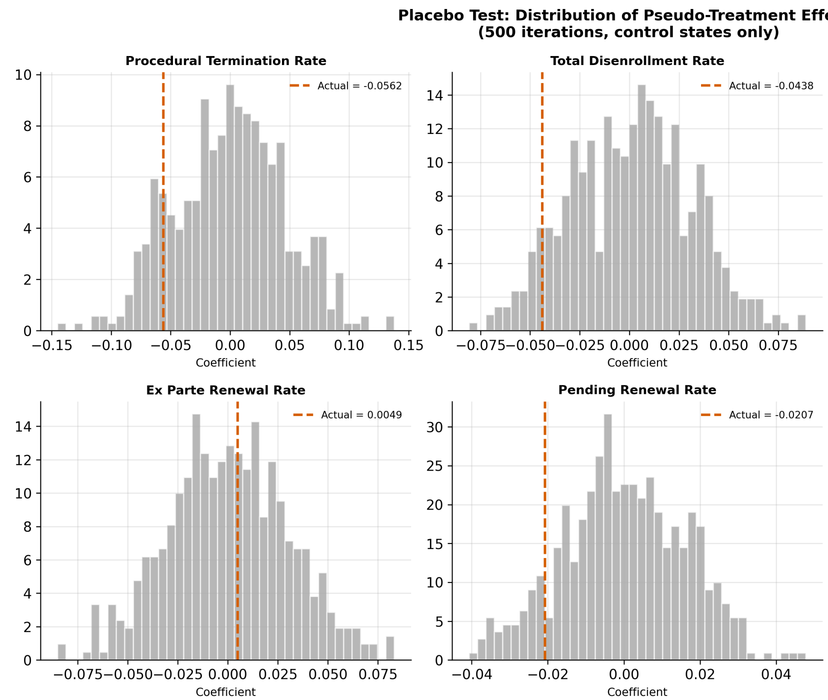

Figure 7: Fig4 Placebo Test

Supplementary Appendix

Federal Renewal-Compliance Enforcement and Medicaid Coverage Retention: Evidence from the 2023 Individual-Level Renewal Shock

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Extended Methods

Sample Construction My analysis panel consists of 918 state-month observations (51 jurisdictions x 18 months). The panel is balanced in terms of the state-month skeleton, though outcome variables are missing for approximately 11 percent of observations, concentrated in the earliest months (February–April 2023) when many states had not yet initiated unwinding-related renewals. I treat these as structurally missing rather than imputing values. The primary analysis restricts to months with non-missing renewal data (N = 764).

Estimating Equations **Two-way fixed effects (TWFE).** My baseline specification is:

$$Y_{st} = \alpha_s + \delta_t + \beta \cdot \text{Treatment}_{st} + \varepsilon_{st}$$

where Y_{st} is the outcome for state s in month t, α_s are state fixed effects, δ_t are month fixed effects, and Treatment_{st} is either the ITT indicator ($\text{Noncompliant}_s \times \text{Post}_t$) or the as-treated indicator (PauseActive_{st}). Standard errors are clustered at the state level.

DID2S (Gardner 2022). In the first stage, I estimate state and month fixed effects using only untreated observations. In the second stage, I regress the residualized outcome on the treatment indicator.

Event study. I estimate:

$$Y_{st} = \alpha_s + \delta_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - T_s^* = k] + \varepsilon_{st}$$

with event window $k \in [-6, +12]$ on the 15 pause/delay states only.

Robustness Procedures **Pre-trend tests.** Joint Wald tests on pre-treatment event study coefficients ($k < -1$).

Placebo tests. 500 iterations randomly assigning pseudo-pause dates from the observed treatment timing distribution to control states.

Rambachan-Roth sensitivity. Confidence intervals under varying degrees of parallel trends violations, calibrated using the maximum pre-period slope change.

Oster bounds. Sensitivity to selection on unobservables, with R-max set to 1.3 times the long regression R-squared (capped at 1.0).

Balanced panel. Restriction to months with at least 90 percent state reporting coverage (N = 705).

Alternative treatment definitions. Absorbing treatment (once paused, always treated); ITT with demographic controls.

Early-pause sensitivity. Re-estimation after excluding the 10 pause states whose pauses began before the September 21, 2023 compliance assessment snapshot.

Multiple-testing adjustment. Bonferroni and Benjamini-Hochberg corrections across the 15 main treatment-effect estimates reported in Table 2 of the main manuscript.

Approximate power / minimum detectable effect. Cluster-randomized-design approximation using the outcome standard deviation, state-level intraclass correlation, average observations per state, and the observed treated/control state counts.

Trimmed event-window sensitivity. Re-estimation of event studies using a narrower window of $k \in [-4, +12]$ to assess the influence of the k = -5 lead.

ITT overlap and decomposition. Tabulation of overlap between ITT-treated and pause states, plus group-specific post-period estimates relative to compliant non-pause states.

Pause spell decomposition. Re-estimation with separate indicators for the first active month of a pause, active pause months after onset, and months after the pause ended.

Pause intensity heterogeneity. Re-estimation allowing the active pause effect to vary by planned pause duration and by whether the pause was narrow in scope or targeted to a subset of beneficiaries.

Why I do not use absorbing-treatment staggered-DiD estimators. Estimators such as Sun-Abraham or Callaway-Sant'Anna are most naturally suited to permanent treatment adoption. My intervention of interest is a temporary pause that turns on and off within some states, so those estimators are not a clean substitute for the main as-treated specification without stronger assumptions about what counts as post-treatment exposure.

Table A1. ITT TWFE Difference-in-Differences Results

Outcome	Coefficient	Std. Error	t-statistic	p-value	N	R-squared
Procedural termination rate	0.037	0.041	0.91	0.37	764	0.537
Total disen-rollment rate	0.012	0.031	0.38	0.70	764	0.602
Renewal completion rate	-0.000	0.033	-0.01	0.99	764	0.587
Ex parte renewal rate	0.028	0.031	0.90	0.37	764	0.791
Pending renewal rate	-0.011	0.017	-0.68	0.50	764	0.592

Notes: ITT treatment defined as noncompliant/still-assessing status (30 states) interacted with post-September 2023 indicator. All specifications include state and month fixed effects. Standard errors clustered at the state level.

Table A2. Rambachan-Roth Sensitivity Analysis

Outcome	M Multiplier	CI Lower	CI Upper	Excludes Zero
Procedural Termination Rate				
	0.0	-0.302	0.095	No
	0.5	-0.413	0.206	No
	1.0	-0.523	0.317	No
Total Disenrollment Rate				
	0.0	-0.157	0.039	No
	0.5	-0.255	0.137	No
	1.0	-0.353	0.235	No
Renewal Completion Rate				
	0.0	-0.045	0.152	No
	0.5	-0.141	0.248	No
	1.0	-0.238	0.345	No
Ex Parte Renewal Rate				
	0.0	-0.160	0.066	No
	0.5	-0.199	0.105	No
	1.0	-0.239	0.145	No

Outcome	M Multiplier	CI Lower	CI Upper	Excludes Zero
Pending Renewal Rate				
	0.0	-0.042	0.054	No
	0.5	-0.055	0.066	No
	1.0	-0.067	0.079	No

Notes: Sensitivity analysis following Rambachan and Roth (2023). M represents the maximum allowed deviation from exact parallel trends, calibrated as multiples of the maximum observed pre-period slope change.

Table A3. Oster (2019) Bounds for Selection on Unobservables

Outcome	Short Reg. Coef.	Short R-sq.	Long Reg. Coef.	Long R-sq.	R-max	Delta-star	Beta-star (delta=1)
Procedural termination rate	0.672	0.016	-0.056	0.539	0.700	-0.024	-0.281
Total disenrollment rate	0.292	0.019	-0.044	0.606	0.788	-0.040	-0.148
Renewal completion rate	0.657	0.013	0.065	0.595	0.774	0.033	-0.117
Ex parte renewal rate	0.601	0.000	0.005	0.791	1.000	0.002	-0.153
Pending renewal rate	0.051	0.001	-0.021	0.594	0.772	-0.087	-0.042

Notes: R-max = 1.3 x R-squared of the long regression (capped at 1.0). Delta-star is the degree of proportional selection on unobservables relative to observables that would drive the coefficient to zero.

Table A4. Mechanisms Test: Procedural Terminations and Renewal Completions

Outcome	Coefficient	Std. Error	p-value	Direction
Procedural termination rate	-0.056	0.038	0.14	-
Procedural disenrollment/due rate	-0.046	0.022	0.04	-
Pending renewal rate	-0.021	0.026	0.42	-

Outcome	Coefficient	Std. Error	p-value	Direction
Total disenrollment rate	-0.044	0.027	0.11	-
Renewal completion rate	0.065	0.032	0.05	+

Notes: All specifications use the as-treated (`pause_active`) treatment with state and month fixed effects. The procedural disenrollment/due rate is the most direct measure of the intervention's intended effect.

Table A5. Children-Affected Heterogeneity (Triple-Difference)

Outcome	ITT Treatment	p-value	ITT x Children Affected	p-value
Procedural termination rate	0.027	0.62	0.015	0.80
Total disenrollment rate	-0.016	0.74	0.045	0.37
Renewal completion rate	0.054	0.24	-0.087	0.08
Ex parte renewal rate	0.039	0.44	-0.017	0.72
Pending renewal rate	-0.038	0.01	0.042	0.11

Notes: Triple-difference specification interacting the ITT treatment with an indicator for whether children were flagged as affected by the compliance issue in the CMS September 2023 assessment.

Table A6. Missingness Sensitivity: Full vs. Restricted Panel

Outcome	Full Panel (N=918)		Restricted Panel (N=764)	
	Coefficient	p-value	Coefficient	p-value
Procedural termination rate	-0.056	0.14	-0.056	0.14
Total disenrollment rate	-0.044	0.11	-0.044	0.11
Renewal completion rate	0.065	0.05	0.065	0.05
Ex parte renewal rate	0.005	0.81	0.005	0.81
Pending renewal rate	-0.021	0.42	-0.021	0.42

Notes: Results are identical because structurally missing observations are automatically dropped from regressions in both specifications.

Table A7. Log-Level Outcomes (As-Treated TWFE)

Outcome	Coefficient	Std. Error	p-value	N
Log procedural disenrollments	-0.507	0.379	0.19	771
Log total disenrollments	-0.243	0.205	0.24	771
Log renewals completed	0.017	0.115	0.88	771

Notes: Semi-elasticity specification using natural log of count outcomes. All specifications include state and month fixed effects with standard errors clustered at the state level.

Table A8. Alternative Treatment Definitions

Outcome	Absorbing Treatment		ITT + Controls	
	Coefficient	p-value	Coefficient	p-value
Procedural termination rate	-0.068	0.25	0.037	0.37
Total disenrollment rate	-0.047	0.29	0.012	0.70
Renewal completion rate	0.087	0.07	-0.000	0.99
Ex parte renewal rate	0.015	0.67	0.028	0.37
Pending renewal rate	-0.040	0.27	-0.011	0.50

Notes: "Absorbing treatment" codes the treatment as one from the first month of pause onward. "ITT + Controls" uses noncompliant status x post-September 2023 with state and month fixed effects.

Table A9. Balanced Panel Results

Outcome	Coefficient	Std. Error	p-value	N
Procedural termination rate	-0.033	0.048	0.50	705
Total disenrollment rate	-0.024	0.029	0.41	705
Renewal completion rate	0.049	0.025	0.06	705
Ex parte renewal rate	-0.015	0.019	0.44	705
Pending renewal rate	-0.025	0.031	0.42	705

Notes: Sample restricted to months with at least 90 percent of states reporting non-missing outcome data.

Table A10. Pre-Trend Test Results

Outcome	N Pre-Period Coefs.	Wald Chi-sq.	Chi-sq. p-value	F-Statistic	F p-value	Reject at 5%?
Procedural termination rate	4	8.94	0.06	2.23	0.07	No
Total disen-rollment rate	4	10.98	0.03	2.74	0.03	Yes
Renewal completion rate	4	12.89	0.01	3.22	0.01	Yes
Ex parte renewal rate	4	0.38	0.98	0.10	0.98	No
Pending renewal rate	4	5.50	0.24	1.38	0.24	No

Notes: Joint Wald test on pre-treatment event study coefficients. Pre-trend rejection for disenrollment and renewal completion rates is driven primarily by the k = -5 coefficient.

Table A11. Placebo Test Results (500 Iterations)

Outcome	Actual Coef.	Placebo Mean	Placebo SD	5th Pctl.	95th Pctl.	Empirical p-value
Procedural termination rate	-0.056	-0.000	0.046	-0.074	0.076	0.25
Total disen-rollment rate	-0.044	0.001	0.030	-0.047	0.049	0.15
Renewal completion rate	0.065	-0.001	0.031	-0.049	0.049	0.04
Ex parte renewal rate	0.005	0.000	0.030	-0.050	0.049	0.88
Pending renewal rate	-0.021	0.000	0.016	-0.027	0.026	0.21

Notes: Placebo test based on 500 iterations with randomly assigned pseudo-pause dates.

Table A12. Sensitivity Excluding Pre-September 2023 Pause States

Outcome	As-Treated TWFE Coef.	p-value	DID2S Coef.	p-value	N
Procedural termination rate	-0.076	0.32	-0.090	0.23	616
Total disenrollment rate	-0.018	0.64	-0.021	0.55	616
Renewal completion rate	0.074	0.11	0.073	0.09	616
Ex parte renewal rate	0.018	0.52	0.015	0.58	616
Pending renewal rate	-0.056	0.22	-0.053	0.24	616

Notes: This sensitivity drops DC, DE, IL, ME, MI, MN, NH, NJ, OK, and SC, leaving 5 pause states. The positive renewal-completion association remains directionally similar but becomes less precise.

Table A13. Multiple-Testing Adjustment Across Main Results

Estimator	Outcome	Coefficient	Raw p-value	Bonferroni p-value	Benjamini-Hochberg p-value
ITT TWFE	Procedural termination rate	0.037	0.37	1.00	0.68
ITT TWFE	Total disenrollment rate	0.012	0.70	1.00	0.87
ITT TWFE	Renewal completion rate	-0.000	0.99	1.00	0.99
ITT TWFE	Ex parte renewal rate	0.028	0.37	1.00	0.68
ITT TWFE	Pending renewal rate	-0.011	0.50	1.00	0.68
As-treated TWFE	Procedural termination rate	-0.056	0.14	1.00	0.53

Estimator	Outcome	Coefficient	Raw p-value	Bonferroni p-value	Benjamini-Hochberg p-value
As-treated TWFE	Total disenrollment rate	-0.044	0.11	1.00	0.53
As-treated TWFE	Renewal completion rate	0.065	0.05	0.75	0.52
As-treated TWFE	Ex parte renewal rate	0.005	0.81	1.00	0.87
As-treated TWFE	Pending renewal rate	-0.021	0.42	1.00	0.68
DID2S	Procedural termination rate	-0.031	0.46	1.00	0.68
DID2S	Total disenrollment rate	-0.027	0.31	1.00	0.68
DID2S	Renewal completion rate	0.064	0.07	1.00	0.52
DID2S	Ex parte renewal rate	0.008	0.77	1.00	0.87
DID2S	Pending renewal rate	-0.038	0.19	1.00	0.58

Notes: Adjustments are applied across the 15 main treatment-effect tests in Table 2. No main estimate survives either Bonferroni or Benjamini-Hochberg correction.

Table A14. Approximate Power And Minimum Detectable Effects

Outcome	Approximate MDE (pp)	Observed As-Treated Effect (pp)	Observed / MDE
Procedural termination rate	12.89	-5.62	-0.44
Total disenrollment rate	8.18	-4.38	-0.54
Renewal completion rate	8.05	6.45	0.80
Ex parte renewal rate	17.00	0.49	0.03
Pending renewal rate	5.33	-2.07	-0.39

Notes: MDE values are approximate 80 percent-power thresholds for the as-treated design using 15 treated states and 36 controls. For renewal completion, the observed effect remains below the approximate detection threshold.

Table A15. Trimmed Event-Window Sensitivity

Outcome	Pre-Trend p-value, [-6,12]	Pre-Trend p-value, [-4,12]	k = 0 Coef. (p), [-6,12]	k = 0 Coef. (p), [-4,12]
Procedural termination rate	0.06	0.47	-0.104 (0.33)	-0.093 (0.37)
Total disenrollment rate	0.03	0.50	-0.059 (0.26)	-0.055 (0.29)
Renewal completion rate	0.01	0.25	0.053 (0.31)	0.049 (0.35)
Ex parte renewal rate	0.98	0.94	-0.047 (0.43)	-0.050 (0.40)
Pending renewal rate	0.24	0.29	0.006 (0.82)	0.006 (0.82)

Notes: Trimming the event-study window removes the k = -5 lead. For renewal completion, this change eliminates formal pre-trend rejection but does not produce a sharp or precisely estimated onset effect.

Table A16. ITT Overlap And Renewal-Completion Decomposition

Panel A. Overlap Between ITT Status And Pause States

Group	Description	N States	States
`itt_pause`	ITT-treated states that also paused	10	CO, DC, DE, IL, KS, ME, MN, NH, NJ, NM
`itt_nonpause`	ITT-treated states without pauses	20	AK, CT, GA, HI, IA, ID, MA, MD, ND, NE, NV, NY, OH, OR, PA, VA, VT, WI, WV, WY
`compliant_pause`	Pause states classified as compliant	5	KY, MI, OK, SC, TX

Panel B. Renewal Completion Rate Decomposition

Group	Description	Coefficient	Std. Error	p-value	N
`itt_pause_post`	ITT-treated states that also paused	0.039	0.053	0.47	764
`itt_nonpause_post`	ITT-treated states without pauses	-0.003	0.037	0.93	764
`compliant_pause_post`	Pause states classified as compliant	0.047	0.051	0.36	764

Notes: Panel B reports group-specific post-period estimates relative to compliant non-pause states in a state- and month-fixed-effects specification. The null ITT estimate reflects incomplete overlap between ITT-treated and pause states and the near-zero post-period change among ITT-treated states that never paused.

Table A17. Pause Spell Decomposition

Outcome	Onset Coef.	p-value	During-Pause Coef.	p-value	Post-Pause Coef.	p-value
Procedural termination rate	-0.117	0.13	-0.066	0.26	-0.042	0.50
Total disenrollment rate	-0.078	0.04	-0.048	0.31	-0.022	0.67
Renewal completion rate	0.107	0.002	0.090	0.08	0.061	0.25
Ex parte renewal rate	-0.012	0.76	0.020	0.58	0.019	0.70
Pending renewal rate	-0.029	0.41	-0.042	0.29	-0.040	0.19

Notes: Specifications include state and month fixed effects with standard errors clustered at the state level. `Onset` denotes the first active month of the pause spell, `During-Pause` denotes active pause months after onset, and `Post-Pause` denotes months after the pause ended in pause states. The observed sample contains 15 onset state-months, 117 during-pause state-months, and 59 post-pause state-months.

Table A18. Pause Intensity Heterogeneity

Outcome	Baseline Pause Coef.	p-value	Narrow/Targeted Pause Differential	p-value	Extra Planned Month Differential	p-value
Procedural termination rate	-0.043	0.27	0.104	0.02	-0.117	0.001
Total disenrollment rate	-0.010	0.67	-0.018	0.74	-0.063	0.01
Renewal completion rate	0.058	0.22	-0.002	0.97	0.018	0.61
Ex parte renewal rate	0.005	0.85	-0.027	0.54	0.021	0.41
Pending renewal rate	-0.048	0.26	0.021	0.52	0.046	0.06

Notes: `Baseline Pause` is the effect for broad, all-beneficiary pauses with a planned delay length of one month. `Narrow/Targeted Pause Differential` captures pauses that were partial in scope or limited to a subset of beneficiaries. `Extra Planned Month Differential` captures each additional planned pause month beyond one. Because every targeted-population pause in this panel is also partial-scope, scope and population are not separately identified and are combined into one differential term.

Table A19. Pause State Profiles

State	Start	End	Realized Active Months	Post-Pause Months In Sample	Planned Delay Months	Scope	Population
CO	2023-09	2024-06	10	1	2	partial	Non-MAGI (long-term care)
DC	2023-06	2024-02	9	5	1	partial	Non-MAGI
DE	2023-05	2024-06	14	1	1	all	All beneficiaries

State	Start	End	Realized Active Months	Post-Pause Months In Sample	Planned Delay Months	Scope	Population
IL	2023-06	2024-06	13	1	1	all	All beneficiaries
KS	2023-09	2024-06	10	1	1	all	All beneficiaries
KY	2023-09	2024-06	10	1	3	all	All beneficiaries and subsets
ME	2023-08	2024-06	11	1	1	all	All beneficiaries
MI	2023-06	2024-06	13	1	1	all	All beneficiaries
MN	2023-06	2023-07	2	12	1	all	All beneficiaries
NH	2023-08	2024-06	11	1	2	partial	Non-MAGI (long-term care)
NJ	2023-06	2023-08	3	11	1	all	All beneficiaries
NM	2023-09	2024-06	10	1	1	all	All beneficiaries
OK	2023-06	2023-07	2	12	1	partial	MAGI
SC	2023-06	2024-06	13	1	2	all	All beneficiaries
TX	2023-10	2023-10	1	9	1	all	All beneficiaries

Notes: Realized active months and post-pause months are measured within the February 2023 to July 2024 analysis window. This table shows why the spell-decomposition design has much stronger leverage for onset and during-pause effects than for post-pause rebound: many states remained paused through June 2024 and contribute only one observed post-pause month.

Figure A1. Individual Event Study Plots

[See the replication workflow]

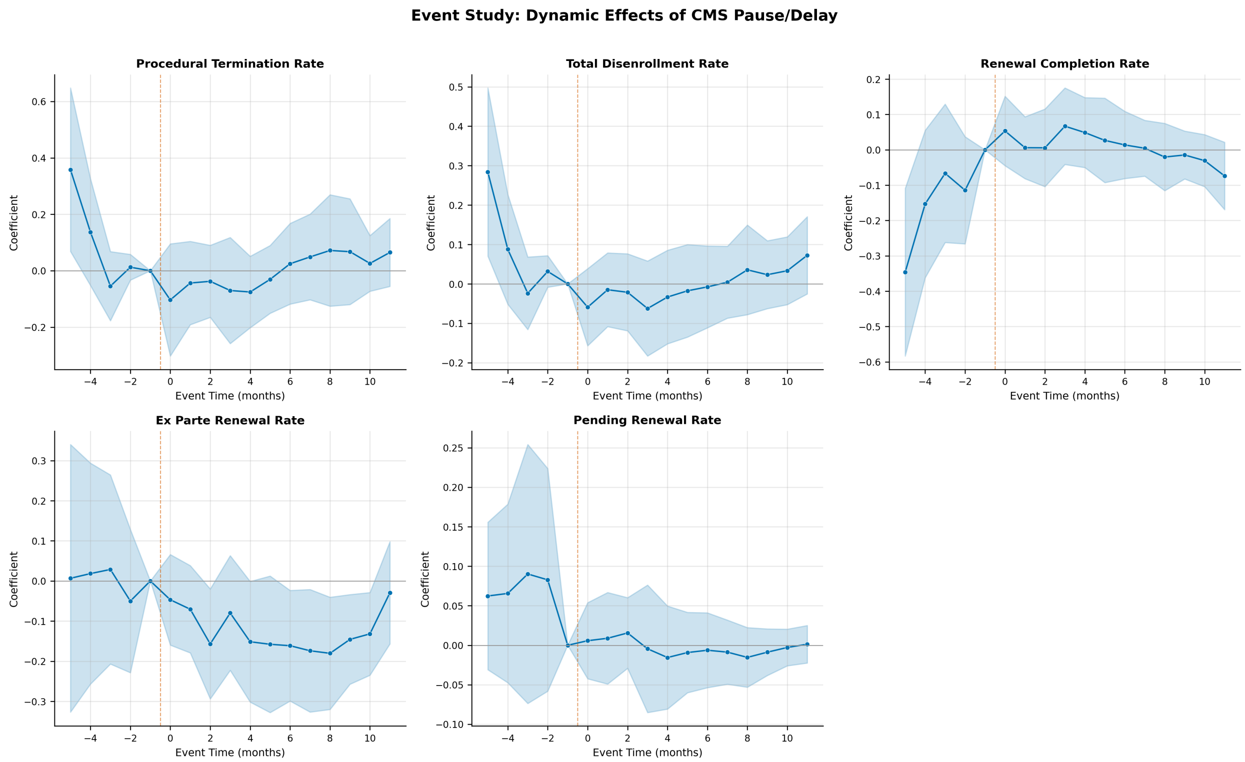

Figure 8: Fig3 Event Study Panel

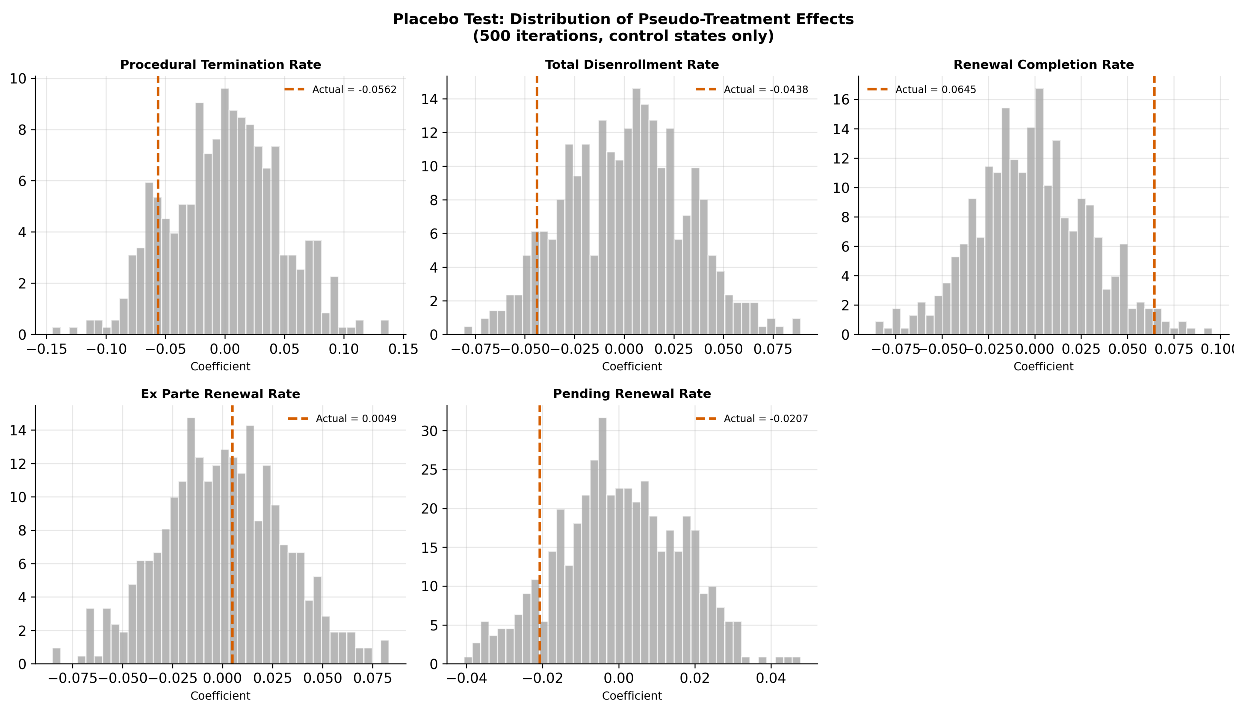

Figure 9: Fig4 Placebo Test

Figure A2. Multi-Panel Event Study Summary

Figure A3. Placebo Test Distributions

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